mitti-netra

MITTI-NETRA soil and structure telemetry node: manufacturing manual

> **Status: design only; nothing built or connected to a mobile network.** PCB: **order-ready: every automatic check passes**. The cellular modem is a plug-in SIM7080G breakout (not on the board). Firmware core host-tested; ESP32 firmware compiles for the ESP32-S3 (arduino-cli, core 2.0.17, 0 warnings; `tools/esp32_build.sh`, binaries in `firmware/release/`), not yet run on hardware.

1. What is made

A 90 x 80 mm solar/USB-charged node: ESP32-S3, TP4056 charger for one protected Li-ion cell, 3.3 V LDO, switched socket for a SIM7080G breakout, ADXL345 tilt/vibration, HX711 strain bridge input, capacitive soil probe input, DS18B20 input. An optional Raspberry Pi/Banana Pi runs `camera/crop_index.py`.

2. Deliverables

Item	Where	State
PCB	`utility10/pcb/release/mitti/` and `mitti-netra/hardware/`	**order-ready: every automatic check passes**
Core library and tests	`mitti-netra/firmware/lib/mn_core.[ch]`, `test/test_mn.c`	`make test` passes (23 checks)
ESP32 skeleton	`mitti-netra/firmware/src/`	compiles for the ESP32-S3 (0 warnings); cellular send is a TODO
Camera analysis	`mitti-netra/camera/crop_index.py`	self-test passes on synthetic scenes

3. Bill of materials

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
U1	ESP32-S3-WROOM-1-N8	ESP32-S3-WROOM-1	ESP32-S3-WROOM-1-N8	C2913198	SMT (JLC)
C37	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C38	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
R42	10k	R_0603_1608Metric	0603 10k 1%	C25804	SMT (JLC)
C39	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
SW2	Reset (EN)	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
SW3	Boot (IO0)	SW_SPST_B3U-1000P	Omron B3U-1000P	C231329	SMT (JLC)
J1	USB-C (charge + program)	USB_C_Receptacle_HRO_TYPE-C-31-M-12	TYPE-C-31-M-12	C165948	SMT (JLC)
R43	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
R44	5.1k	R_0603_1608Metric	0603 5.1k 1%	C23186	SMT (JLC)
J2	Solar panel 5-6 V	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
D1	SS14	D_SMA	SS14	C84343	SMT (JLC)
D2	SS14	D_SMA	SS14	C84343	SMT (JLC)
C1	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
U4	TP4056	HSOP-8-1EP_3.9x4.9mm_P1.27mm_EP2.41x3.1mm	TPOWER TP4056	C382139	SMT (JLC)
R1	2.2k	R_0603_1608Metric	0603 2.2k 1%	C4190	SMT (JLC)
R16	6.2k	R_0603_1608Metric	UNI-ROYAL 0603WAF6201T5E	C4260	SMT (JLC)
TH1	NTC 10k B3950	R_0805_2012Metric	FH CMFB103F3950FANT	C51597	SMT (JLC)
C2	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
D3	Red	LED_0603_1608Metric	Hubei KENTO KT-0603R red (Everlight 19-217 was out of stock)	C2286	SMT (JLC)
R2	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
D4	Green	LED_0603_1608Metric	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	SMT (JLC)
R3	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
J3	Li-ion cell (protected!)	JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical	JST B2B-PH-K-S(LF)(SN)	C131337	hand/THT
U3	AP2112K-3.3	SOT-23-5	AP2112K-3.3TRG1	C51118	SMT (JLC)
C3	1u	C_0603_1608Metric	0603 1uF X7R	C15849	SMT (JLC)
C4	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
R4	330k	R_0603_1608Metric	UNI-ROYAL 0603WAF3303T5E	C23137	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
R5	330k	R_0603_1608Metric	UNI-ROYAL 0603WAF3303T5E	C23137	SMT (JLC)
C5	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
Q1	AO3401A	SOT-23	AO3401A	C15127	SMT (JLC)
R6	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
Q2	AO3400A	SOT-23	AO3400A	C20917	SMT (JLC)
R7	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
R8	100k	R_0603_1608Metric	0603 100k 1%	C25803	SMT (JLC)
C6	100u/35V	CP_Elec_6.3x7.7	Lelon VZH101M1VTR-0607	C176683	SMT (JLC)
C7	22u/10V	C_0805_2012Metric	Samsung CL21A226MPQNNNE	C29277	SMT (JLC)
J6	Cellular modem (SIM7080G breakout)	PinHeader_2x03_P2.54mm_Vertical	HCTL PZ254-2-03-Z-8.5	C2894964	hand/THT
U5	ADXL345	LGA-14_3x5mm_P0.8mm_LayoutBorder1x6y	ADXL345BCCZ-RL7	C9667	SMT (JLC)
C8	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C9	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
R9	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
R10	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
U2	HX711	SOIC-16_3.9x9.9mm_P1.27mm	HX711	C43656	SMT (JLC)
R11	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
C10	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C11	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C12	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
C13	10u	C_0805_2012Metric	0805 10uF X5R	C15850	SMT (JLC)
R12	100	R_0603_1608Metric	UNI-ROYAL 0603WAF1000T5E	C22775	SMT (JLC)
C14	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)

Ref	Value	Package	Part (manufacturer)	LCSC	Assembly
C15	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
J4	Strain-gauge bridge E+ E- A+ A-	JST_PH_B4B-PH-K_1x04_P2.00mm_Vertical	JST B4B-PH-K-S(LF)(SN)	C131334	hand/THT
J5	Soil probe (capacitive, 3.3 V)	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT
J7	DS18B20 temp	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S	C131339	hand/THT
R13	4.7k	R_0603_1608Metric	0603 4.7k 1%	C23162	SMT (JLC)
R15	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
C16	100n	C_0603_1608Metric	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	SMT (JLC)
D5	Green	LED_0603_1608Metric	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	SMT (JLC)
R14	1k	R_0603_1608Metric	0603 1k 1%	C21190	SMT (JLC)
J8	UART flash/debug (USB-serial adapter)	JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical	JST B3B-PH-K-S(LF)(SN)	C131339	hand/THT

Bought-in

SIM7080G (LTE-M/NB-IoT) breakout board with nano-SIM holder and antenna; protected 18650 cell (or protected pouch cell) with JST-PH 2 lead; 5-6 V solar panel (2 W) with JST-PH 2 lead; capacitive soil moisture probe (3.3 V, analog) with JST-PH 3 lead; DS18B20 waterproof probe; strain gauges (350 ohm, GF about 2) with bridge completion; IP65 enclosure with cable glands; a SIM and data plan.

4. Make the board

1. Upload `utility10/pcb/release/mitti/1_fabrication/mitti_netra_gerbers.zip` to the PCB fab (2 layers, 1.6 mm, ENIG or lead-free HASL, green mask, white silk; 1 oz is fine). 2. For assembly upload `2_assembly/mitti_netra_JLC_BOM.csv` and `mitti_netra_JLC_CPL.csv`; check every polarity in the preview (diodes, electrolytic capacitor, USB-C, ESP32 module pin 1). 3. Through-hole connectors and pin sockets are listed with LCSC numbers but are normally hand-soldered. 4. General rules, tolerances and inspection: `manuals/STANDARD_PROCEDURES.md` sections A and B. The ADXL345 is an LGA-14 part: use the fab's reflow assembly; do not hand-solder it.

5. Program and configure

Arduino-ESP32 core 2.x: `mitti-netra/firmware/src/main_mitti.cpp` plus `firmware/lib/mn_core.c`. Set the placeholders: soil calibration counts (dry air, water), strain excitation/gauge factor/bridge type, tilt and strain

thresholds. Install: tilt the node into its final position, power up once: that boot stores the baseline (tilt angles and strain zero) in RTC memory; a power loss loses the baseline.

6. Test (see `mitti-netra/docs/TEST_PLAN.md`)

Charge LEDs; TP2 3.3 V; deep-sleep current; modem switch toggles VBAT_M; ADXL345 ID 0xE5; tilt plate 10 degrees reads 10 +-0.5; HX711 shunt-resistor check; 7-day outdoor run.

7. Limits

No cell protection on the board (use protected cells). LDO stops below about 3.55 V on the cell. Strain numbers are conversions, not a calibrated instrument. Carrier APN, bands and a server are not provided. Not safety-rated.